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# pg_routing.tcl
# Description  : Power/Ground (PG) connection and mesh generation script.
# Tool         : Synopsys ICC2 (or similar implementation tool)
# Requirements : Core area defined, VDD/VSS rails present.
# Notes        :
#   - Creates temporary M1 routing blockages around hard macros
#   - Builds PG straps on M8 and M7
#   - Connects standard cell rails on M1
#   - Adds stacked PG vias from M8 down to M1
#   - Cleans temporary blockages and saves block

# 1. Connect existing PG nets (rails/straps)
puts "\n[INFO] Connecting existing PG nets (automatic)...\n"
connect_pg_net -automatic

# 2. Create M1 routing blockage around hard macros
set my_macros [get_flat_cells * -filter "is_hard_macro == true"]
set halo 0.5   ;# Halo around macros in routing units (e.g., microns)

if { [sizeof_collection $my_macros] == 0 } {
  puts "\n[WARNING] No hard macros found. Skipping M1 routing blockage
creation.\n"
} else {
  puts "\n[INFO] Creating M1 routing blockages around hard macros...\n"
  foreach_in_collection a $my_macros {
    set macro_llx [lindex [get_attribute $a bbox_ll] 0]
    set macro_lly [lindex [get_attribute $a bbox_ll] 1]
    set macro_urx [lindex [get_attribute $a bbox_ur] 0]
    set macro_ury [lindex [get_attribute $a bbox_ur] 1]

    set macro_llx_new [expr {$macro_llx - $halo}]
    set macro_lly_new [expr {$macro_lly - $halo}]
    set macro_urx_new [expr {$macro_urx + $halo}]
    set macro_ury_new [expr {$macro_ury + $halo}]

    set macro_bbox_new [list [list $macro_llx_new $macro_lly_new] \
                [list $macro_urx_new $macro_ury_new]]
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      puts "[INFO] create_routing_blockage -name_prefix M1_PG -layers {M1} 
-boundary $macro_bbox_new"
      create_routing_blockage -name_prefix M1_PG -layers {M1} -boundary 
$macro_bbox_new
   }
}

# 3. Create power straps on M8 (horizontal)
puts "\n[INFO] Creating PG mesh pattern and strategy for M8 straps...\n"

create_pg_mesh_pattern strap_pattern_M8 \
   -layers {{horizontal_layer: M8 \
          width: 8 \
          spacing: minimum} \
         {trim: false \
          offset: 3}}

set_pg_strategy M8_straps -design_boundary \
   -pattern {{name: strap_pattern_M8} {nets: VDD VSS}} \
   -extension {stop: design_boundary_and_generate_pins}

compile_pg -strategies M8_straps

# 4. Create power straps on M7 (vertical)
puts "\n[INFO] Creating PG mesh pattern and strategy for M7 straps...\n"

create_pg_mesh_pattern strap_pattern_M7 \
   -layers {{vertical_layer: M7 \
          width: 7 \
          spacing: minimum} \
         {trim: false \
          offset: 3}}

set_pg_strategy M7_straps -design_boundary \
   -pattern {{name: strap_pattern_M7} {nets: VDD VSS}} \
   -extension {stop: design_boundary_and_generate_pins}
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compile_pg -strategies M7_straps

# 5. Create standard cell PG rails on M1
puts "\n[INFO] Creating standard cell PG rails on M1...\n"

create_pg_std_cell_conn_pattern std_rail \
   -layers {M1} \
   -check_std_cell_drc true \
   -mark_as_follow_pin true

set_pg_strategy rail_stra -core \
   -pattern {{name: std_rail} {nets: VDD VSS}} \
   -extension {stop: core_boundary}

compile_pg -strategies rail_stra

# 6. Add stacked PG vias from M8 down to M1
puts "\n[INFO] Adding stacked PG vias from M8 to M1 over core area...\n"

set bbox [get_attribute [get_core_area] bbox]

create_pg_vias \
   -nets {VDD VSS} \
   -from_layers M8 \
   -to_layers M1 \
   -within_bbox $bbox \
   -drc no_check

# 7. Remove temporary M1 routing blockages
puts "\n[INFO] Removing temporary M1 routing blockages (M1_PG)...\n"

remove_routing_blockages M1_PG

# 8. Save block after PG routing
puts "\n[INFO] Saving block state after PG routing...\n"
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save_block -as ${design}_pg_routing_done

# 9. Optional PG verification (uncomment as needed)
# check_pg_drc
# check_pg_connectivity

puts "\n[INFO] PG routing script completed successfully.\n"
```